

## SARANYA POOLA

Sr. Java Full Stack Developer

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### PROFESSIONAL SUMMARY

- **Senior Java Full Stack Developer** with over **10 years of experience** building enterprise-grade solutions for critical systems in **financial services, healthcare, insurance, manufacturing, and banking** sectors.
- Specialized in **object-oriented programming (OOP) and design principles**, creating scalable, maintainable software architectures leveraging Java 17, Spring Boot, and Microservices patterns.
- Highly skilled in **frontend development** with React JS (17+) and Angular (versions 8-17), delivering dynamic, user-centric interfaces with component-based architecture.
- Built and integrated **RESTful and SOAP-based web services** utilizing Spring MVC, JAX-RS, and JAX-WS; incorporated GraphQL to provide adaptable data querying capabilities.
- Extensive expertise in **database design and data persistence** across MySQL, PostgreSQL, Oracle, MongoDB, and Elasticsearch platforms.
- Architected **event-driven systems** leveraging Apache Kafka and RabbitMQ for asynchronous messaging and real-time data processing workflows.
- Established **automated CI/CD workflows** using Jenkins, GitLab, and Maven, streamlining build automation, testing protocols, and multi-environment deployments.
- Deployed **containerized microservices** with Docker and managed orchestration through Kubernetes, guaranteeing system portability and continuous availability.
- Secured application layers using **Spring Security and OAuth 2.0** protocols, implementing role-based access controls and ensuring adherence to PCI-DSS and HIPAA regulations.
- Practiced **test-driven development (TDD)** with JUnit, Mockito, and Postman; implemented **behavior-driven development (BDD)** methodologies using Cucumber frameworks.
- Managed **full-cycle Agile delivery** including sprint planning, daily standups, and retrospectives, partnering with product stakeholders and quality assurance teams.
- Proven track record in **modernizing legacy monolithic applications** to microservice architectures, achieving 30-50% improvements in deployment velocity and system scalability.
- Conducted **technical code reviews**, provided mentorship to junior developers, and championed coding best practices, enhancing codebase quality across geographically distributed teams.
- Proficient in **application performance optimization**, including database query tuning, JVM diagnostics, and memory utilization analysis to maximize system efficiency.
- Deployed **comprehensive logging and observability** frameworks (Log4J, Slf4J, CloudWatch, Splunk, SonarQube) enabling real-time monitoring and maintaining code quality metrics.
- Experienced in **UI/UX development**, converting design prototypes into interactive web applications using HTML5, CSS3, Bootstrap 5, and Tailwind CSS.
- Versatile in **multi-cloud and hybrid environments** with expertise in multi-tenant system design, following enterprise-level DevSecOps methodologies.
- Acknowledged for **analytical problem-solving**, effective stakeholder communication, and converting business requirements into robust technical implementations.
- Dedicated to **ongoing professional development**, consistently integrating emerging Java versions, framework enhancements, and cloud-native best practices to deliver cutting-edge, reliable systems.

### EDUCATION

- Bachelor's from Jawaharlal Nehru Technological University, Hyderabad. Aug, 2009 – May, 2013

### SKILL SET

|                              |                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | Java 8 / 11 / 17, JavaScript (ES6 +), TypeScript, Python, SQL, PL/SQL, Scala              |
| <b>Frameworks</b>            | Spring, Spring Boot, Spring MVC, Spring Cloud, Spring Security, Hibernate, JPA, React JS, |

|                                  |                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|
|                                  | Angular 8–17, Node JS, Struts                                                                              |
| <b>Web Technologies</b>          | HTML5, CSS3, Bootstrap 4 / 5, Tailwind CSS, AJAX, JSON, XML, REST API, GraphQL, JSP, JSTL                  |
| <b>Database&amp; Tools</b>       | MySQL, PostgreSQL, Oracle 11g / 12c, MongoDB, DynamoDB, Cassandra, Elasticsearch, pgAdmin, MySQL Workbench |
| <b>Messaging &amp; Streaming</b> | Apache Kafka, RabbitMQ, IBM MQ, Active MQ, JMS                                                             |
| <b>Web Application Servers</b>   | Oracle WebLogic 10.x / 9.x, Apache Tomcat 8.x / 7.x / 6.x, IBM WebSphere 8.x / 6.x                         |
| <b>Project Methodologies</b>     | Agile (Scrum / Kanban), Waterfall, JIRA, Confluence, Asana, Slack, Microsoft Teams                         |
| <b>Testing &amp; Quality</b>     | JUnit, Mockito, Selenium, Postman, Swagger, Jasmine, Karma, Cucumber, SonarQube, Splunk, PowerMockito      |
| <b>Operating Systems</b>         | Windows, Linux, UNIX, Mac OS                                                                               |
| <b>Cloud Platforms</b>           | AWS (EC2, ECS, EBS, S3, RDS, Lambda, CloudWatch, IAM), Azure, GCP                                          |
| <b>Security &amp; Auth</b>       | OAuth 2.0, JWT, SSO, SSL / TLS, Role-Based Access Control                                                  |

## PROFESSIONAL EXPERIENCE

Citi Bank, New York, NY

Mar 2024 – Present

**Sr. Java Full Stack Developer**

**Project name:** Global Credit Card Payment Processing System

**Project Description:** Architected and optimized Citi's enterprise-wide payment processing system, enabling real-time credit and debit transaction handling while integrating advanced fraud detection capabilities and maintaining strict adherence to PCI-DSS requirements and internal compliance protocols.

**Responsibilities:**

- Engaged in **requirements gathering, functional reviews, and sprint planning sessions**, working alongside product owners, business analysts, and quality assurance teams to convert business requirements into detailed technical designs for international payment processing modules.
- Developed scalable microservices using **Java 21**, Spring Boot, and Spring Cloud, ensuring system maintainability.
- By applying layered architecture patterns and modular design principles to guarantee maintainability and stability across distributed systems.
- Built **RESTful API services** managing payment authorization workflows, transaction routing logic, and merchant settlement processes while implementing idempotency controls and secure communication protocols between internal and third-party services.
- Constructed **data persistence layers** leveraging Spring Data JPA and Hibernate, fine-tuning entity relationships, caching mechanisms, and connection pool configurations for high-throughput transactional operations.
- Developed **dynamic, responsive front-end applications** using React JS paired with Redux for centralized state management, enabling real-time display of transaction records, compliance audit trails, and fraud detection dashboards.
- Incorporated **AWS cloud services** including Lambda, RDS, S3, and CloudWatch to build serverless execution workflows, unified logging infrastructure, and auto-scaling capabilities for compute-intensive background tasks.
- Deployed **containerized microservices** through Docker and managed orchestration using Kubernetes (EKS) clusters, ensuring continuous uptime and streamlined deployment processes across geographically distributed environments.
- Set up **Spring Cloud Config Server** for unified configuration management and deployed Eureka Service Registry to enable dynamic service discovery across the microservices ecosystem.
- Built **event-driven processing frameworks** utilizing Apache Kafka and RabbitMQ to orchestrate payment notifications, general ledger synchronization, and reconciliation events across decentralized architectures.

- Implemented **API security measures** using Spring Security, OAuth 2.0, and JWT token-based authentication to protect endpoints and enforce granular authorization controls throughout frontend and backend layers.
- Partnered with **enterprise architects** to design high-availability and fault-resilient service architectures, facilitating seamless connectivity with upstream payment authorization platforms and downstream analytics systems.
- Leveraged **Log4j, Splunk, and AWS CloudWatch** for aggregated application logging, exception tracking, and continuous monitoring of system performance and operational health metrics.
- Streamlined **CI/CD automation** using Jenkins, Maven, and Git, incorporating static code analysis and unit test coverage verification prior to production releases.
- Engaged in **peer code reviews, collaborative programming, and architectural discussions** to uphold clean code standards and comply with organizational development guidelines.
- Coordinated with **QA professionals** to design extensive regression, integration, and smoke test frameworks using JUnit, Mockito, and Postman for thorough validation.
- Created dynamic front-end applications using **React JS** and Redux for real-time transaction dashboards and fraud alerts.
- Developed reusable **React** component libraries to streamline UI development and ensure consistency.
- Generated **comprehensive API documentation** via Swagger/OpenAPI specifications and managed endpoint versioning to facilitate easier integration and testing for cross-functional teams.
- Supported **DevOps operations** in environment setup, load balancer optimization, and version control branching approaches to enable seamless continuous delivery workflows.
- Contributed to **daily Scrum ceremonies and sprint reviews** to address impediments, optimize team processes, and enhance delivery velocity across distributed development groups.
- Facilitated **user acceptance testing cycles** and coordinated with release management for production rollouts, contingency planning, and post-deployment verification activities.
- Created **comprehensive technical documentation** including system architecture blueprints, integration flow diagrams, and deployment procedures to promote knowledge sharing and transparency across international development teams.

**Environment:** *Java 21, Spring Boot, Spring Cloud, React, ReactJS, MySQL, Elasticsearch, Kafka, RabbitMQ, Docker, Kubernetes (EKS), AWS Lambda, S3, RDS, CloudWatch, Maven, Jenkins, Git, Swagger, JUnit, Mockito, Log4j, Splunk*

**Mastercard, New York, NY**

**Jan 2022 –Feb 2024**

**Sr. Java Full Stack Developer**

**Project name:** Enterprise Network Monitoring and Security Platform

**Project Description:** Contributed to building Mastercard's global real-time monitoring and analytics solution engineered to observe, analyze, and protect network operations across international financial infrastructures. The platform delivered continuous event tracking, automated incident management, and regulatory compliance reporting capabilities through the integration of diverse data streams within a unified cloud-based architecture hosted on Microsoft Azure.

**Responsibilities:**

- Participated in requirements gathering sessions with network operations and information security teams to establish performance benchmarks, event detection criteria, and notification protocols consistent with Mastercard's operational guidelines.
- Architected and built highly scalable **Spring Boot** microservices leveraging **Java 17** to process high-volume event capture, data transformation, and consolidation across geographically distributed infrastructure.
- Developed an event-driven system architecture utilizing **Apache Kafka**, facilitating real-time data streaming and microservice messaging for monitoring and alerting capabilities.
- Designed **RESTful API** interfaces to integrate with third-party incident management platforms and internal data processing workflows while maintaining secure and dependable data exchange.
- Engineered interactive frontend dashboards using **Angular 15** with **TypeScript, RxJS**, and **Bootstrap 5** to deliver real-time visualization of security alerts, system anomalies, and infrastructure performance indicators.
- Incorporated **Elasticsearch** as the core search and analytics engine for event cataloging, pattern recognition, and historical compliance log queries.
- Containerized and deployed microservices using Docker, managing orchestration and auto-scaling via **Azure Kubernetes Service (AKS)** to ensure system resilience and business continuity.

- Established automated **CI/CD** workflows through Azure DevOps and Jenkins, integrating code quality assessments, automated testing, and version control mechanisms throughout the software delivery process.
- Configured Spring Security, **JWT**, and **OAuth 2.0** protocols to provide comprehensive authentication, session management, and hierarchical authorization controls across user groups and service endpoints.
- Employed Spring Data **JPA** for database operations and optimized query performance to efficiently manage high-volume event datasets within **Azure SQL Database** environments.
- Engaged with solution architects and lead engineers to establish service contracts, define microservice responsibilities, and standardize interaction protocols across distributed system components.
- Partnered with **DevOps** engineers to deploy comprehensive observability solutions using **Azure Monitor**, Log Analytics, and Application Insights for real-time alerting and operational transparency.
- Implemented standardized error handling and logging strategies using **Log4J** and **SLF4J frameworks**, enhancing diagnostic capabilities across the microservices landscape and accelerating issue resolution.
- Developed Python scripts for data analysis, automating transaction reconciliation, and generating compliance reports. And utilized **Python** with **Pandas** and **Numpy** for fraud pattern analysis and real-time data processing.
- Built Python-based automation tools for **CI/CD** pipeline optimization and environment management. Also Integrated **Python** machine learning models with Java microservices using **REST APIs** for predictive fraud detection.
- Employed **Python** for log analysis and monitoring, enhancing system observability with Splunk and CloudWatch. Also created **Python** utilities for database migration, performance testing, and batch processing tasks.
- Leveraged **Django** to develop internal admin panels for transaction monitoring and system configuration.
- Actively contributed to Agile Scrum rituals, including sprint planning sessions, backlog grooming, and daily standups, promoting inter-team collaboration and milestone achievement.
- Facilitated thorough code review processes via **GitHub** pull requests, enforcing Mastercard's secure development standards and maintaining codebase quality and consistency.
- Developed extensive unit and integration test suites using **JUnit 5**, **Mockito**, and Postman to verify **REST API** functionality and validate cross-layer service behavior.
- Produced and updated **Swagger/OpenAPI** specifications for all exposed APIs to streamline testing procedures, validation workflows, and external system integration activities.
- Offered mentorship and technical coaching to junior team members, delivering training on microservice design patterns, **Azure** cloud infrastructure, and **Agile** development methodologies.
- Engaged in internal technical forums, delivering presentations on enhancing system monitoring, refining **CI/CD** processes, and implementing **Azure-native** performance tuning strategies.

**Environment:** Java 17, Spring Boot, Python, Spring Cloud, Angular 14, Apache Kafka, Elasticsearch, PostgreSQL, AWS RDS, Docker, Kubernetes (EKS), Jenkins, Maven, Git, Swagger, JUnit, Mockito, CloudWatch, JWT, OAuth 2.0

**MEDICAL GUARDIAN — Philadelphia, PA**

**Aug**

**2019 - Dec 2021**

**Java Full Stack Developer**

**Project Title:** Health and Human Services Management Platform

**Responsibilities:**

- Worked closely with **healthcare specialists, product stakeholders, and system architects** to elicit and evaluate requirements, ensuring full compliance with HIPAA standards and protected health information (PHI) protocols.
- Architected and implemented **microservices using Python frameworks** (Django and Flask) supporting patient registration workflows, notification systems, and medical device data integration.
- Developed and connected **RESTful API services** with wearable health monitoring devices and external emergency response providers to enable secure, instantaneous data exchange.
- Created **dynamic user interface components** leveraging Angular 8–9, TypeScript, and Bootstrap 4/5 to deliver dashboards presenting patient health metrics, system alerts, and critical event notifications.
- Established **JWT-based authentication mechanisms** and role-specific access policies using Django Security and OAuth 2.0 protocols for protected user identity management.
- Leveraged **AWS cloud infrastructure** (Lambda, S3, RDS) for serverless computation, encrypted data storage, and device telemetry management; deployed Apache Kafka for asynchronous event distribution and live data transmission across monitoring and analytical services.
- Administered **relational and NoSQL databases** including PostgreSQL, MongoDB, DynamoDB, and Cosmos DB, fine-tuning queries to achieve optimal system responsiveness.

- Streamlined **build automation and release processes** through Jenkins, GitHub, and Azure DevOps, maintaining reliable continuous integration and deployment operations.
- Developed and ran **comprehensive test cases** using pytest, unittest, and Postman to ensure service dependability and data integrity across all system layers.
- Created **API documentation** via Swagger/OpenAPI specifications to simplify testing workflows and facilitate seamless integration with external platforms; containerized microservices using Docker and orchestrated deployments on Kubernetes (AKS) environments to support scalability and system reliability.
- Engaged in **Agile Scrum practices**, participating in sprint planning sessions, daily synchronization meetings, and team retrospectives; tracked application health through Splunk, AWS CloudWatch, and SonarQube, establishing preventive monitoring and performance enhancement measures.

**Environment:** Java 17, Spring Boot 3.x, Spring Cloud, Angular 15, TypeScript, Bootstrap 5, Apache Kafka, Elasticsearch, Azure SQL Database, Azure Kubernetes Service (AKS), Azure Monitor, Application Insights, Azure DevOps, Docker, Jenkins, Maven, Git, Swagger/OpenAPI, JUnit 5, Mockito, JWT, OAuth 2.0, Log4j, SLF4j, Agile/Scrum

**METLIFE Cary, NC**

**Feb 2018 - Aug 2019**

**Java Developer**

**Project Title:** Insurance Claim Management System

**Responsibilities:**

- Engaged in **requirements elicitation workshops** with business analysts to establish claims processing workflows, policy validation rules, and external system integration specifications.
- Built and supported **Spring MVC and Spring Boot RESTful services** facilitating claim origination, modification, and resolution operations across diverse organizational divisions.
- Modernized **multiple legacy Struts-based applications** to Spring Boot architecture, enhancing code maintainability and supporting distributed microservice deployment strategies.
- Engineered **data persistence frameworks** utilizing Hibernate ORM and Spring Data JPA integrated with Oracle 11g, defining entity mappings and performance-optimized database procedures.
- Configured **Spring Security mechanisms** for authorization management and protected confidential claims data through authentication protocols, access restrictions, and encrypted communication channels.
- Developed **client-side interfaces** using JSP, JSTL, HTML5, and Bootstrap 4, enhancing claims submission workflows and improving user interface responsiveness.
- Integrated with **SOAP and REST web services** from external providers for policy retrieval, payment processor connectivity, and third-party claims adjudication capabilities.
- Established **automated CI/CD pipelines** leveraging Jenkins and Maven, coordinating with quality assurance and operations teams for environment promotions and production deployments.
- Deployed **standardized exception management** and application logging using Log4j to enhance error tracking capabilities and facilitate runtime system diagnostics.
- Created **reusable code libraries and utility modules** to expedite bulk claims processing operations and automate status synchronization activities.
- Designed and performed **unit and integration testing** using JUnit 5 and Mockito, verifying claims lifecycle API functionality and maintaining regression test coverage.
- Participated in **Agile Scrum workflows**, including sprint planning meetings, daily synchronization calls, and retrospective sessions to sustain delivery momentum and resolve impediments efficiently.
- Investigated and remediated **production system issues**, delivering emergency patches and liaising with support personnel to minimize business impact on claims operations.
- Produced **detailed technical artifacts**, including API documentation, specification guides, and database schema illustrations to support system transparency and enable future platform improvements.

**Environment:** Java 8, Spring Boot, Spring MVC, Hibernate, Spring Data JPA, Oracle 11g, JSP, JSTL, Bootstrap 4, Struts, Jenkins, Maven, JUnit 5, Mockito, Spring Security, Log4j

**Dell Technologies, Chennai, India.**

**June 2013**

**—Sep 2016**

**Java Developer**

**Project Title:** Product Inventory and Order Management System

**Responsibilities:**

- Engaged in **requirements analysis and design review sessions** to comprehend inventory management processes and product distribution logic utilized across multiple manufacturing operations.
- Built and maintained **server-side application modules** using J2EE, Servlets, and JSP technologies for orchestrating purchase orders, stock level management, and vendor relationship workflows.
- Developed **business process logic and transactional operations** leveraging Spring Framework and Hibernate, guaranteeing data accuracy and operational consistency throughout system components.
- Created and optimized **PL/SQL queries, database procedures, and automated triggers** within Oracle 11g to efficiently process large-scale transactional datasets.
- Designed **reusable frontend components** utilizing HTML5, CSS3, JavaScript, and Bootstrap, delivering adaptive user interfaces for order monitoring and inventory status visualization.
- Incorporated **reporting capabilities** using JasperReports to produce dynamic analytical dashboards for executive leadership and warehouse operations personnel.
- Established **Log4J and Splunk configurations** for unified logging infrastructure, facilitating streamlined troubleshooting and continuous performance assessment.
- Deployed and administered **applications on Oracle WebLogic and Apache Tomcat platforms**, managing server configurations and deployment automation procedures.
- Implemented **comprehensive error-handling mechanisms** and service-layer validation rules to ensure system reliability across interconnected application modules.
- Worked with **interdisciplinary teams** to optimize order fulfillment workflows and minimize manual processing through scripted automation solutions.
- Coordinated with **quality assurance professionals** for functional validation, regression testing protocols, and load performance evaluation using JUnit and Selenium frameworks.
- Contributed to **collaborative code review processes**, maintaining compliance with organizational development standards and version management practices in SVN repositories.
- Delivered **operational production support** through log analysis, system fault diagnosis, and deployment of corrective updates to resolve post-deployment challenges.
- Created **comprehensive system documentation**, including architectural blueprints, integration workflow diagrams, and database design specifications for knowledge preservation and regulatory adherence.

**Environment:** Java 7, J2EE, Spring Framework, Hibernate, Oracle 11g, JSP, Servlets, HTML5, CSS3, JavaScript, Bootstrap, Log4J, Splunk, WebLogic, Tomcat, JasperReports, JUnit, Selenium, SVN